

# **CS-GY 6923: Lecture 3**

## **Regularization + Bayesian Perspective**

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- The first written HW will be posted this weekend.
- 10% extra credit will be given if solutions are typewritten using LaTeX, Markdown, etc.
- All solutions must be written independently.
- Reminder: Lab 02 due September 24.

## Model selection:

- Train models  $f_{\theta_1}^{(1)}, \dots, f_{\theta_q}^{(q)}$  independently on training data to find optimal parameters  $\theta_1^*, \dots, \theta_q^*$ .
- Check loss  $L_{test}(f^{(1)}, \theta_1^*), \dots, L_{test}(f^{(q)}, \theta_q^*)$  on test data.
- Select model with lowest test loss.

Can be used for arbitrary sets of models. Often used when you are not sure how “complex” your model should be for the data, and want to find a sweet spot between a good fit, and not overfitting.

# Model selection 1: train-test paradigm

**More reasonable approach:** Evaluate model on fresh test data which was not used during training.

## Test/train split:

- Given data set  $(\mathbf{X}, \mathbf{y})$ , split into two sets  $(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$  and  $(\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}})$ .
- Train  $q$  models  $f^{(1)}, \dots, f^{(q)}$  of varying complexity by finding parameters which minimize the loss on  $(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$ .
- Evaluate loss of each trained model on  $(\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}})$ .
- Pick model with lowest test loss.

Sometimes you will see the term **validation set** instead of test set. Sometimes there will be both: use validation set for choosing the model, and test set for getting a final performance measure.



Is “test error” the end goal though? Don’t we care about “future” error?

**Intuition:** Models which perform better on the test set will **generalize** better to future data.

**Goal:** Introduce a little bit of formalism to better understand what this means. What is “future” data?

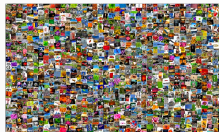
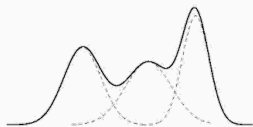
# Statistical learning model

## Statistical Learning Model:

- Assume each data example is randomly drawn from some distribution  $(\mathbf{x}, y) \sim \mathcal{D}$ .

E.g.  $x_1, \dots, x_d$  are Gaussian random variables with parameters

$$\mu_1, \sigma_1, \dots, \mu_d, \sigma_d.$$



This is not really a simplifying assumption! The distribution could be arbitrarily complicated.

## Statistical Learning Model:

- Assume each data example is randomly drawn from some distribution  $(\mathbf{x}, y) \sim \mathcal{D}$ .
- Define the **Risk** of a model/parameters:

$$R(f, \theta) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [L(f(\mathbf{x}, \theta), y)]$$

here  $L$  is our loss function (e.g.  $L(z, y) = |z - y|$  or  $L(z, y) = (z - y)^2$ ).

**Ultimate Goal:** Find model  $f \in \{f^{(1)}, \dots, f^{(q)}\}$  and parameter vector  $\theta$  to minimize the  $R(f, \theta)$ .

- **(Population) Risk:**

$$R(f, \theta) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [L(f(\mathbf{x}, \theta), y)]$$

- **Empirical Risk:** Draw  $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n) \sim \mathcal{D}$

$$R_E(f, \theta) = \frac{1}{n} \sum_{i=1}^n L(f(\mathbf{x}_i, \theta), y_i)$$

For any fixed model  $f$  and parameters  $\theta$ ,

$$\mathbb{E}[R_E(f, \theta)] = R(f, \theta).$$

Only true if  $f$  and  $\theta$  are chosen *without looking at the data used to compute the empirical risk*.

# Model selection

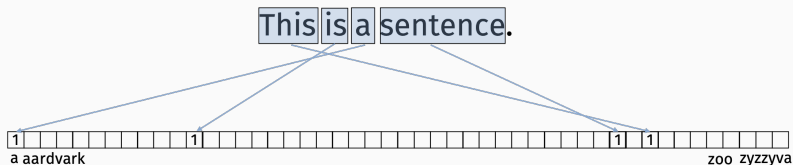
- Train  $q$  models  $(f^{(1)}, \theta_1^*), \dots, (f^{(q)}, \theta_q^*)$ .
- For each model, compute empirical risk  $R_E(f^{(i)}, \theta_i^*)$  using test data.
- Since we assume our original dataset was drawn independently from  $\mathcal{D}$ , so is the random test subset.

No matter how our models were trained or how complex they are,  $R_E(f^{(i)}, \theta_i^*)$  is an unbiased estimate of the true risk  $R(f^{(i)}, \theta_i^*)$  for every  $i$ . Can use it to distinguish between models.

## Model selection 2: bag of words

### bag-of-words models and n-grams

Common way to represent documents (emails, webpages, books) as numerical data. The ultimate example of 1-hot encoding.

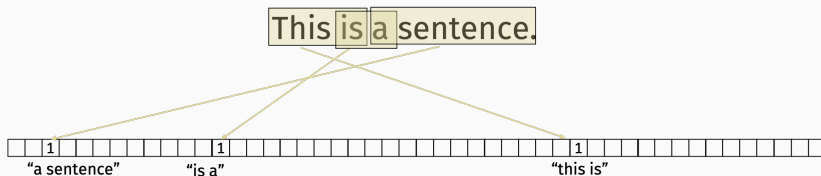


**bag-of-words**

# Model selection example

## bag-of-words models and n-grams

Common way to represent documents (emails, webpages, books) as numerical data. The ultimate example of 1-hot encoding.



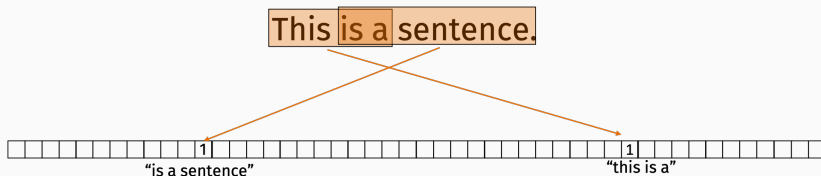
**bi-grams**



# Model selection example

## bag-of-words models and n-grams

Common way to represent documents (emails, webpages, books) as numerical data. The ultimate example of 1-hot encoding.



**tri-grams**

# Model selection example

## Models of increasing order:

- Model  $f_{\theta_1}^{(1)}$ : spam filter that looks at **single words**.
- Model  $f_{\theta_2}^{(2)}$ : spam filter that looks at **bi-grams**.
- Model  $f_{\theta_3}^{(3)}$ : spam filter that looks at **tri-grams**.
- ...

“interest”

“low interest”

“low interest loan”

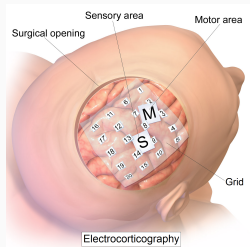
Increased length of **n-gram** means more expressive power.

**Will be very relevant in our lab on generative language models!**

# Model selection 3: autoregressive models

## Electrocorticography ECoG (Lab 02):

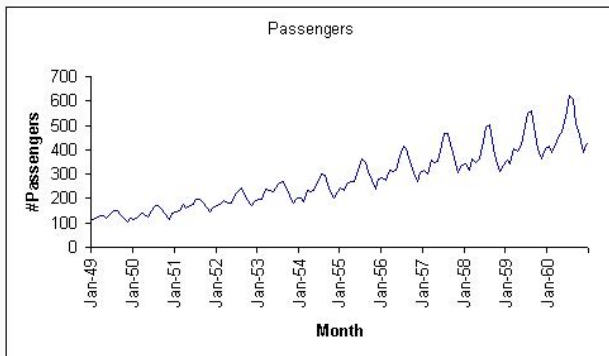
- Implant grid of electrodes on surface of the brain to measure electrical activity in different regions.



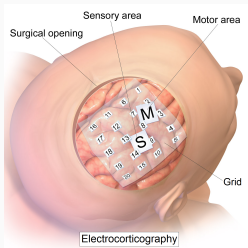
- Predict hand motion based on ECoG measurements.
- **Model order:** predict movement at time  $t$  using brain signals at time  $t, t - 1, \dots, t - q$  for varying values of  $q$ .

# Autoregressive model

Predicting time  $t$  based on a linear function of the signals at time  $t, t - 1, \dots, t - q$  is not the same as fitting a line to the time series. It's much more expressive.



## Electrocorticography ECoG lab:



**First lab where computation actually matters (solving regression problems with  $\sim 40k$  examples,  $\sim 1500$  features)**

Makes sense to test and debug code using a subset of the data.

**Slight caveat: The train-test paradigm is typically not how machine learning or scientific discovery works in practice!**

## Typical workflow:

- Train a class of models.
- Test.
- Adjust class of models.
- Test.
- Adjust class of models.
- Cont...

Final model implicitly depends on test set because performance on the test set guided how we changed our model.

## Popularity of ML benchmarks and competitions leads to adaptivity at a massive scale.

| 11 Active Competitions                                                            |                                                                                                                                                                                                       |                            |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
|  | <b>Deepfake Detection Challenge</b><br>Identify videos with facial or voice manipulations<br><i>Featured</i> · Code Competition · 2 months to go · video data, online video                           | \$1,000,000<br>1,595 teams |
|  | <b>Google QUEST Q&amp;A Labeling</b><br>Improving automated understanding of complex question answer content<br><i>Featured</i> · Code Competition · 19 hours to go · text data, nlp                  | \$25,000<br>1,559 teams    |
|  | <b>Real or Not? NLP with Disaster Tweets</b><br>Predict which Tweets are about real disasters and which ones are not<br><i>Getting Started</i> · Ongoing · text data, binary classification           | \$10,000<br>2,657 teams    |
|  | <b>Bengali.AI Handwritten Grapheme Classification</b><br>Classify the components of handwritten Bengali<br><i>Research</i> · Code Competition · a month to go · multiclass classification, image data | \$10,000<br>1,194 teams    |

Kaggle (various competitions)



14,197,122 images, 21841 synsets indexed

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Imagenet (image classification and categorization)

## Is adaptivity a problem? Does it lead to over-fitting? How much? How can we prevent it?

### REPORT

## The reusable holdout: Preserving validity in adaptive data analysis

Cynthia Dwork<sup>1,\*</sup>, Vitaly Feldman<sup>2,\*</sup>, Moritz Hardt<sup>3,\*</sup>, Toniann Pitassi<sup>4,\*</sup>, Omer Reingold<sup>5,\*</sup>, Aaron Roth<sup>6,\*</sup>

+ See all authors and affiliations

*Science* 07 Aug 2015:  
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## Do ImageNet Classifiers Generalize to ImageNet?

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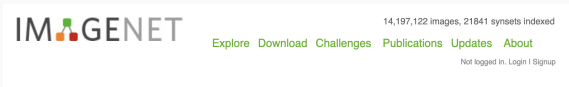
### Abstract

We build new test sets for the CIFAR-10 and ImageNet datasets. Both benchmarks have been the focus of intense research for almost a decade, raising the danger of overfitting to excessively re-used test sets. By closely following the original dataset creation processes, we test to what extent current classification models generalize to new data. We evaluate a broad range of models and find accuracy drops of 3% – 15% on CIFAR-10 and 11% – 14% on ImageNet. However, accuracy gains on the original test sets translate to larger gains on the new test sets. Our results suggest that the accuracy drops are not caused by adaptivity, but by the models' inability to generalize to slightly "harder" images than those found in the original test sets.

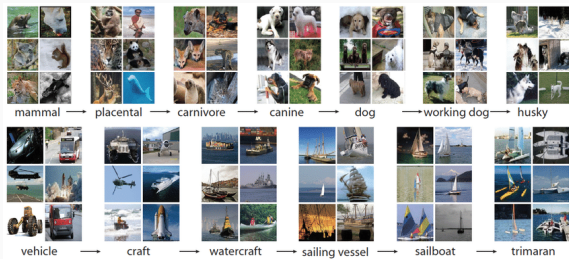
12 Jun 2019



# Imagenet dataset

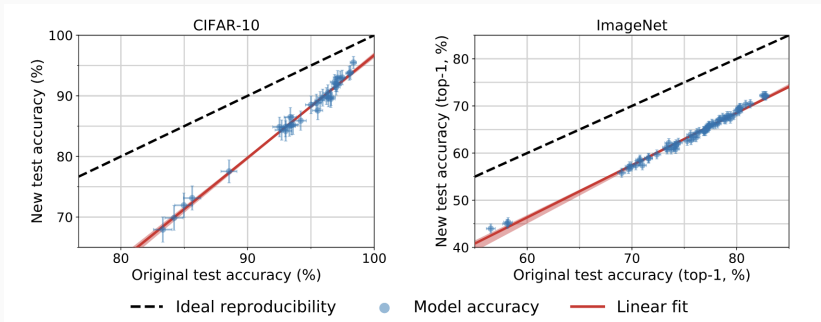


Collected by Fei-Fei Li's group at Stanford in 2006ish and labeled using Amazon Mechanical Turk.



We now have neural network models that can solve these classification problems with  $> 95\%$  accuracy.

## Do ImageNet Classifiers Generalized to ImageNet?



Interestingly, when comparing popular vision models on “fresh” data, while performance dropped across the board, the relative rank of model performance did not change significantly.

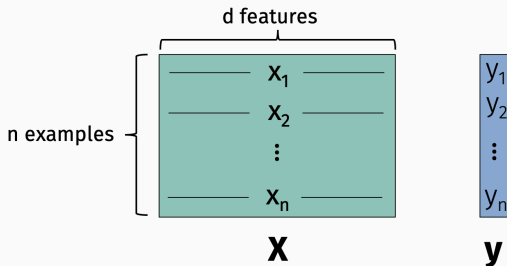
**Regularization:**  
**another way of controlling model complexity**

## Over-parameterized models

- In the model selection examples we discussed last class, we had full control over the complexity of the model: could range from underfitting to overfitting.
- In practice, you often don't have this freedom. Even the most basic model might lead to overfitting.

# Over-parameterized models

**Example:** Linear regression model where  $d \geq n$ .

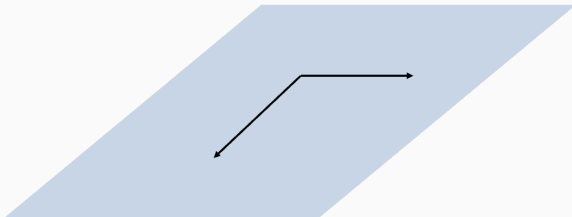


Can (almost) always find  $\beta$  so that  $\mathbf{X}\beta = \mathbf{y}$  exactly.

Why?

# High dimensional linear models

**Claim:** For almost all sets of  $n$ , length  $n$  vectors  $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}$ , we can write any vector  $\mathbf{y}$  as a linear combination of these vectors.



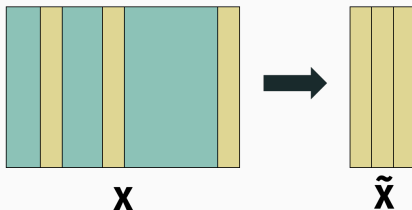
I.e., can find some coefficients so that  
$$\beta_1 \mathbf{x}^{(1)} + \dots + \beta_d \mathbf{x}^{(d)} = \mathbf{X}\boldsymbol{\beta} = \mathbf{y}.$$

## Zero train loss

- We will discuss some models later in the class where zero training loss is not necessarily a bad sign:  $k$ -nearest neighbors (the inductive bias of  $k$ -NN is local similarity), some neural nets (e.g., some empirical studies on double descent).
- Typically however it will be a sign of overfitting, as in the polynomial regression example.

# Feature selection

Select some subset of  $\ll n$  features to use in model:



**Filter method:** Compute some metric for each feature, and select features with highest score.

- Example: compute loss or  $R^2$  value when each feature in  $\mathbf{X}$  is used in single variate regression.

**Any potential limitations of this approach?**



**Exhaustive approach:** Pick best subset of  $q$  features. Train  $\binom{d}{q}$  models.

**Faster approach:** Greedily select  $q$  features.

## Stepwise Regression:

- **Forward:** Step 1: pick single feature that gives lowest loss.  
Step  $k$ : pick feature that when combined with previous  $k - 1$  chosen features gives lowest loss.
- **Backward:** Start with all of the features. Greedily eliminate those which have least impact on model performance.

**Feature selection deserves more than two slides, but we won't go into too much more detail!**

**Regularization:** Discourage overfitting by adding a regularization penalty to the loss minimization problem.

$$\min_{\beta} [L(\beta) + \textit{Reg}(\beta)].$$

## Alternative approach

**Regularization:** Discourage overfitting by adding a regularization penalty to the loss minimization problem.

$$\min_{\beta} [L(\beta) + \text{Reg}(\beta)].$$

**Example:** Least squares regression:  $L(\beta) = \|\mathbf{X}\beta - \mathbf{y}\|_2^2$ .

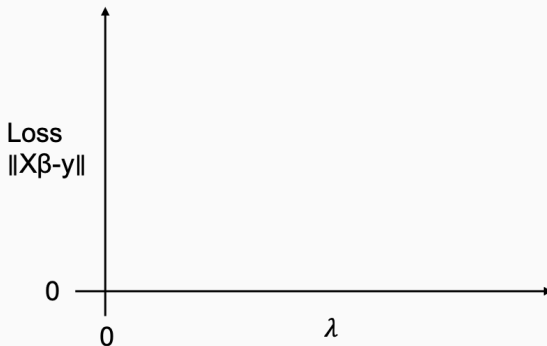
- Ridge regression ( $\ell_2$ ):  $\text{Reg}(\beta) = \lambda \|\beta\|_2^2$
- LASSO (least absolute shrinkage and selection operator) ( $\ell_1$ ):  
 $\text{Reg}(\beta) = \lambda \|\beta\|_1$
- Elastic net:  $\text{Reg}(\beta) = \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2$

Note that  $\arg \min_{\beta} [L(\beta) + \text{Reg}(\beta)] \neq \arg \min_{\beta} [L(\beta)]$

## Ridge regularization: perspective 1

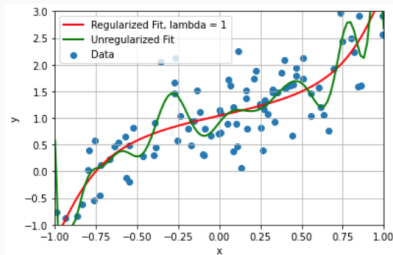
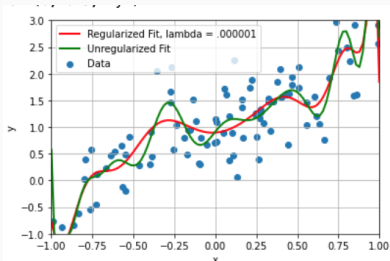
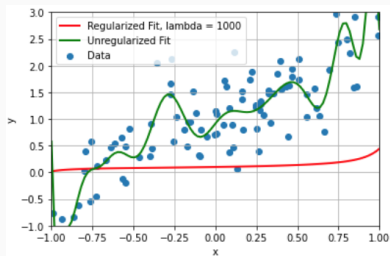
Ridge regression:  $\min_{\beta} (\|\mathbf{X}\beta - \mathbf{y}\|_2^2 + \lambda\|\beta\|_2^2)$ .

- As  $\lambda \rightarrow \infty$ , we expect  $\|\beta\|_2^2 \rightarrow 0$  and  $\|\mathbf{X}\beta - \mathbf{y}\|_2^2 \rightarrow \|\mathbf{y}\|_2^2$ .
- By choosing different values of  $\lambda$  we have models of varying accuracy/complexity.



# Polynomial examples

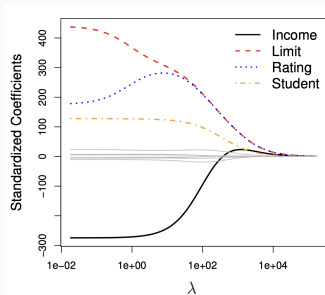
Fit degree 20 polynomial with varying levels of regularization.



## Ridge regularization: perspective 2

Ridge regression:  $\min_{\beta} (\|\mathbf{X}\beta - \mathbf{y}\|_2^2 + \lambda\|\beta\|_2^2)$ .

- As  $\lambda \rightarrow \infty$ , we expect  $\|\beta\|_2^2 \rightarrow 0$  and  $\|\mathbf{X}\beta - \mathbf{y}\|_2^2 \rightarrow \|\mathbf{y}\|_2^2$ .
- Feature selection methods attempt to set many coordinates in  $\beta$  to 0. Ridge regularizations encourages coordinates to be close to zero.



## Ridge regularization

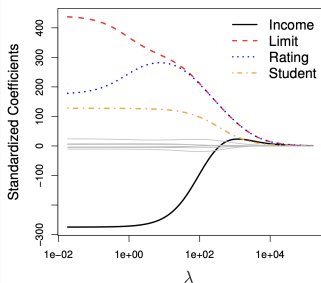
How do we minimize:  $L_R(\beta) = \|\mathbf{X}\beta - \mathbf{y}\|_2^2 + \lambda\|\beta\|_2^2$ ?



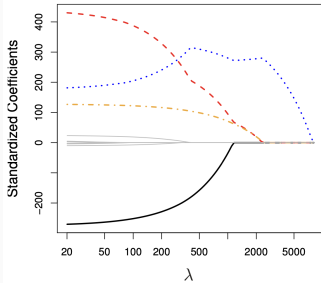
# Lasso regularization

Lasso regularization:  $\min_{\beta} \|\mathbf{X}\beta - \mathbf{y}\|_2^2 + \lambda\|\beta\|_1$ .

- As  $\lambda \rightarrow \infty$ , we expect  $\|\beta\|_1 \rightarrow 0$  and  $\|\mathbf{X}\beta - \mathbf{y}\|_2^2 \rightarrow \|\mathbf{y}\|_2^2$ .
- Typically encourages subset of  $\beta_i$ 's to go to zero, in contrast to ridge regularization.



Ridge regularization



Lasso Regularization

## Pros:

- Simpler, more interpretable model.
- More intuitive reduction in model order.

## Cons:

- No closed form solution because  $\|\beta\|_1$  is not differentiable.
- Can be solved with iterative methods, but generally not as quickly as ridge regression.

## Notes:

- Model selection/cross validation used to choose optimal scaling  $\lambda$  on  $\lambda\|\beta\|_2^2$  or  $\lambda\|\beta\|_1$ .
- Often grid search for best parameters is performed in “log space”. E.g. consider  $[\lambda_1, \dots, \lambda_q] = 1.5^{[-4, -3, -2, -1, -0, 1, 2, 3, 4]}$ .
- Regularization methods are not invariant to data scaling.  
Typically when using regularization we mean center and scale columns to have unit variance.

**The bayesian/probabilistic modeling perspective**

# Classification setup

- **Data Examples:**  $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$
- **Target:**  $y_1, \dots, y_n \in \{0, 1, \dots, q - 1\}$  when there are  $q$  classes.
  - Binary Classification:  $q = 2$ , so each  $y_i \in \{0, 1\}$ .
  - Multi-class Classification:  $q > 2$ .<sup>1</sup>

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<sup>1</sup>Note that there is also multi-label classification where each data example may belong to more than one class.

# Classification examples

- Medical diagnosis from MRI: 2 classes.
- MNIST digits: 10 classes.
- Full Optical Character Recognition: 100s of classes.
- ImageNet challenge: 21,000 classes.

**Running example today: Email Spam Classification.**

# Classification

Classification can (and often is) solved using the same **loss-minimization framework** we saw for regression.

We won't see that today! We're going to use classification as a window into another way of thinking about machine learning.

Will give an interesting new justifications for tools like regularization. Will also give us an approach for generative ML.

**Rest of Today:** ML from a **Probabilistic Modeling/Bayesian Perspective**.

# Probabilistic modeling

In a Bayesian or Probabilistic approach to machine learning we always start by conjecturing a

## probabilistic model

that plausibly could have generated our data.

- The model guides how we make predictions.
- The model typically has unknown parameters  $\vec{\theta}$  and we try to find the most reasonable parameters based on **observed** data (more on this later in lecture).

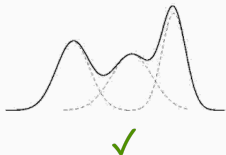


# Naive bayes classifier

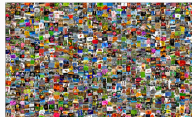
## Goal:

- Build a probabilistic model for a binary classification problem.
- Estimate parameters of the model.
- From the model derive a classification rule for future predictions (the **Naive Bayes Classifier**).

Typically we try to keep things simple!

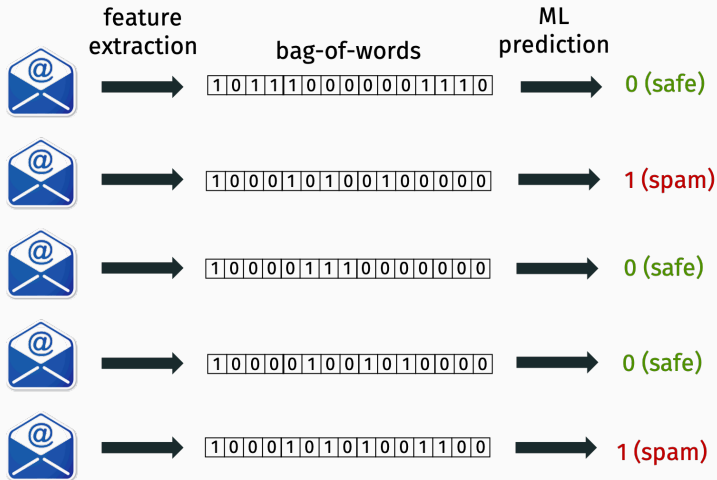


X



X

# Spam prediction



Both target labels and data vectors are binary.

# Email model

Let's create a model that generates spam and non-spam emails.

**Observation:** Since bag-of-words features don't care about word order, our model does not need to either.

- Common approach: assign a probability  $p_i \in [0, 1]$  to word  $i$ .  
Set  $\mathbf{x}_i = 1$  with probability  $p_i$ ,  $\mathbf{x}_i = 0$  with probability  $1 - p_i$ .

$p_{\text{the}} =$

$p_{\text{calendar}} =$

$p_{\text{toothbrush}} =$

# Email model

Search: toothbrush

Active

Mail Conversations Spaces

From Any time Has attachment To Exclude Promotions Is unread Advanced

1-17 of 17

|                          |                          |                          |                     |       |                                                                                                                    |             |  |  |  |          |
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| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Amazon.com          | Inbox | Hooray, it's Prime Day! - Philips Sonicare <b>Toothbrush</b> by Philips Sonicare https://www.amazon.com/g...       |             |  |  |  | 7/11/23  |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | UNHCR Canada        | Inbox | Last chance to give more than a gift - gift of <b>toothbrushes</b> , toothpaste, towels, soap and jerry cans to... |             |  |  |  | 12/22/22 |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Uber Eats           | Inbox | Running low on essentials? - Get \$12 off a \$25+ order of hand soap, <b>toothbrushes</b> , and more.              |             |  |  |  | 7/7/22   |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Amazon.ca           | Inbox | Deals have landed for Cyber Monday! - Sonicare Electric <b>Toothbrushes</b> & Air... https://www.amazon...         |             |  |  |  | 11/27/21 |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Simon Fraser Dental | Inbox | Referral Prize - complimentary electric <b>toothbrush</b> . Refer A Friend & Receive A FREE Oral-B Vitality Re...  |             |  |  |  | 4/28/18  |

# Email model

How can we make this model richer when we take spam into account?

- Different words tend to be more or less frequent in spam or regular emails.

Not Spam

$$p_{\text{won}} =$$

$$p_{\$} =$$

$$p_{\text{student}} =$$

Spam

$$p_{\text{won}} =$$

$$p_{\$} =$$

$$p_{\text{student}} =$$

# Probabilistic model for email

**Probabilistic model** for (bag-of-words, label) pair  $(\mathbf{x}, y)$ :

- Set  $y = 0$  with probability  $p_0$ ,  $y = 1$  with probability  $p_1 = 1 - p_0$ .
  - $p_0$  is probability an email is not spam (e.g. 99%).
  - $p_1$  is probability an email is spam (e.g. 1%).
- If  $y = 0$ , for each  $i$ , set  $x_i = 1$  with prob.  $p_{i0}$ .
- If  $y = 1$ , for each  $i$ , set  $x_i = 1$  with prob.  $p_{i1}$ .

**Unknown model parameters:**

- $p_0, p_1$ ,
- $p_{10}, p_{20}, \dots, p_{d0}$ , one for each of the  $d$  vocabulary words.
- $p_{11}, p_{21}, \dots, p_{d1}$ , one for each of the  $d$  vocabulary words.

**How would you estimate these parameters?**

## Reasonable way to set parameters:

- Set  $p_0$  and  $p_1$  to the empirical fraction of not spam/spam emails.
- For each word  $i$ , set  $p_{i0}$  to the empirical probability word  $i$  appears in a non-spam email.
- For each word  $i$ , set  $p_{i1}$  to the empirical probability word  $i$  appears in a spam email.

**Estimating these parameters from previous data examples is the only “training” we will do.**

Done with modeling  
on to prediction



# Probability review

- **Probability:**  $p(x)$  – the probability event  $x$  happens.
- **Joint probability:**  $p(x,y)$  – the probability that event  $x$  and event  $y$  happen.
- **Conditional Probability**  $p(x | y)$  – the probability  $x$  happens given that  $y$  happens.

$$p(x|y) =$$

## Bayes theorem/rule

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

**Proof:**

# Classification rule

Given unlabeled input ( $\mathbf{w}$ , —), choose the label  $y \in \{0, 1\}$  which is most likely given the data. Recall  $\mathbf{w} = [0, 0, 1, \dots, 1, 0]$ .

**Classification rule:** maximum a posterior (MAP) estimate.

**Step 1. Compute:**

- $p(y = 0 \mid \mathbf{w})$ : prob.  $y = 0$  given observed data vector  $\mathbf{w}$ .
- $p(y = 1 \mid \mathbf{w})$ : prob.  $y = 1$  given observed data vector  $\mathbf{w}$ .

**Step 2. Output:** 0 or 1 depending on which probability is larger.

$p(y = 0 \mid \mathbf{w})$  and  $p(y = 1 \mid \mathbf{w})$  are called **posterior** probabilities.

# Evaluating the posterior

How to compute the posterior? **Bayes rule!**

$$p(y = 0 \mid \mathbf{w}) = \frac{p(\mathbf{w} \mid y = 0)p(y = 0)}{p(\mathbf{w})} \quad (1)$$

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}} \quad (2)$$

- **Prior:** Probability in class 0 prior to seeing any data.
- **Posterior:** Probability in class 0 after seeing the data.

# Evaluating the posterior

Goal is to determine which is larger:

$$p(y = 0 \mid \mathbf{w}) = \frac{p(\mathbf{w} \mid y = 0)p(y = 0)}{p(\mathbf{w})} \quad \text{vs.}$$

$$p(y = 1 \mid \mathbf{w}) = \frac{p(\mathbf{w} \mid y = 1)p(y = 1)}{p(\mathbf{w})}$$

- We can ignore the evidence  $p(\mathbf{w})$  since it is the same for both sides!
- $p(y = 0)$  and  $p(y = 1)$  already known (computed from training data). These are our computed parameters  $p_0, p_1$ .
- $p(\mathbf{w} \mid y = 0) = ?$   $p(\mathbf{w} \mid y = 1) = ?$

## Evaluating the posterior

Consider the example  $\mathbf{w} = [0, 1, 1, 0, 0, 0, 1, 0]$ .

Recall that, under our model, index  $i$  is 1 with probability  $p_{i0}$  if we are not spam, and 1 with probability  $p_{i1}$  if we are spam .

$$p(\mathbf{w} \mid y = 0) =$$

$$p(\mathbf{w} \mid y = 1) =$$

## Final Naive Bayes Classifier

**Training/Modeling:** Use existing data to compute:

- $p_0 = p(y = 0), p_1 = p(y = 1)$
- For all  $i$  compute:
  - $p_{i0} = p(x_i = 1 \mid y = 0)$  and  $(1 - p_{i0}) = p(x_i = 0 \mid y = 0)$
  - $p_{i1} = p(x_i = 1 \mid y = 1)$  and  $(1 - p_{i1}) = p(x_i = 0 \mid y = 1)$

**Prediction:**

- For new input  $\mathbf{w}$ :
  - Compute  $p(\mathbf{w} \mid y = 0) = \prod_i p(w_i \mid y = 0)$
  - Compute  $p(\mathbf{w} \mid y = 1) = \prod_i p(w_i \mid y = 1)$
- Return

$$\arg \max [p(\mathbf{w} \mid y = 0) \cdot p(y = 0), p(\mathbf{w} \mid y = 1) \cdot p(y = 1)].$$

## Other applications of the bayesian perspective



# Bayesian regression

The Bayesian view offers an interesting alternative perspective on many machine learning techniques.

**Example:** Linear Regression.

**Probabilistic model:**

$$y = \langle \mathbf{x}, \boldsymbol{\beta} \rangle + \eta$$

where the  $\eta$  drawn from  $N(0, \sigma^2)$  is **random Gaussian noise**.



$$Pr(\eta = z) \sim$$

The symbol  $\sim$  means “is proportional to”.

# Gaussian distribution refresher

**Names for the same thing:** Normal distribution, Gaussian distribution, bell curve.

Parameterized by **mean**  $\mu$  and **variance**  $\sigma^2$ .



$\eta$  is a continuous random variable, so it has a probability density function  $p(\eta)$  with  $\int_{-\infty}^{\infty} p(\eta) d\eta = 1$

$$p(\eta) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\eta-\mu}{\sigma}\right)^2}$$

# Gaussian distribution refresher

The important thing to remember is that the the PDF falls off exponentially as we move further from the mean.



The normalizing constant in front  $1/2$ , etc. don't matter so much.

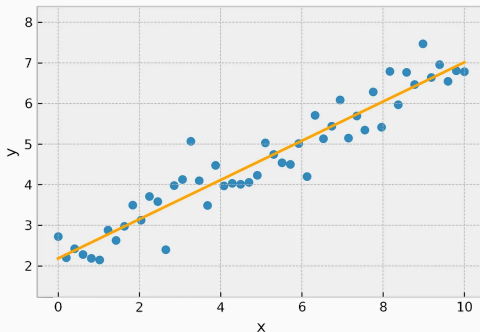
# Illustration

**Example:** Linear Regression.

**Probabilistic model:**

$$y = \langle \mathbf{x}, \boldsymbol{\beta} \rangle + \eta$$

where the  $\eta$  drawn from  $N(0, \sigma^2)$  is **random Gaussian noise**. The noise is independent for different inputs  $\mathbf{x}_1, \dots, \mathbf{x}_n$ .



## How should we select $\beta$ for our model?

Also use a Bayesian approach!

Choose  $\beta$  to maximize:

$$\text{posterior} = \Pr(\beta \mid \mathbf{X}, \mathbf{y}) = \frac{\Pr(\mathbf{X}, \mathbf{y} \mid \beta) \Pr(\beta)}{\Pr(\mathbf{X}, \mathbf{y})} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}.$$

In this case, we don't have a prior – no values of  $\beta$  are inherently more likely than others.

Choose  $\beta$  to maximize just the likelihood:

$$\frac{\Pr(\mathbf{X}, \mathbf{y} \mid \beta) \Pr(\beta)}{\Pr(\mathbf{X}, \mathbf{y})} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}.$$

This is called the **maximum likelihood estimate**.

## Fixed design linear regression

Often we think of  $\mathbf{X}$  as fixed and deterministic, and only  $\mathbf{y}$  is generated at random in the model. This is called the fixed design setting. Can also consider a randomized design setting, but it is slightly more complicated.

In the fixed design setting our task of maximizing  $\Pr(\mathbf{X}, \mathbf{y} \mid \beta)$  simplifies to maximizing

$$\max_{\beta} \Pr(\mathbf{y} \mid \beta)$$

# Maximum likelihood estimate

**Data:**

$$\mathbf{X} = \begin{bmatrix} \text{—} & \mathbf{x}_1 & \text{—} \\ \text{—} & \mathbf{x}_2 & \text{—} \\ & \vdots & \\ \text{—} & \mathbf{x}_n & \text{—} \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

**Model:**  $y_i = \langle \mathbf{x}_i, \boldsymbol{\beta} \rangle + \eta_i$  where  $p(\eta_i = z) \sim e^{-z^2/2\sigma^2}$  and  $\eta_1, \dots, \eta_n$  are independent.

$$\Pr(\mathbf{y} \mid \boldsymbol{\beta}) \sim$$

# Log likelihood

Easier to work with the **log likelihood**:

$$\begin{aligned}\arg \max_{\beta} \Pr(\mathbf{X}, \mathbf{y} \mid \beta) &= \arg \max_{\beta} \prod_{i=1}^n e^{-(y_i - \langle \mathbf{x}_i, \beta \rangle)^2 / 2\sigma^2} \\ &= \arg \max_{\beta} \log \left( \prod_{i=1}^n e^{-(y_i - \langle \mathbf{x}_i, \beta \rangle)^2 / 2\sigma^2} \right) \\ &= \arg \max_{\beta} \sum_{i=1}^n -(y_i - \langle \mathbf{x}_i, \beta \rangle)^2 / 2\sigma^2 \\ &= \arg \min_{\beta} \sum_{i=1}^n (y_i - \langle \mathbf{x}_i, \beta \rangle)^2.\end{aligned}$$



**Conclusion:** Choose  $\beta$  to minimize:

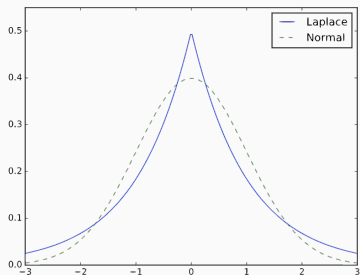
$$\sum_{i=1}^n (y_i - \langle \mathbf{x}_i, \beta \rangle)^2 = \|\mathbf{y} - \mathbf{X}\beta\|_2^2.$$

**This is a completely different justification for squared loss!**

Minimizing the  $\ell_2$  loss is optimal in a certain sense when you assume your data follows a linear model with i.i.d. Gaussian noise.

# Bayesian regression

If we had modeled our noise  $\eta$  as Laplace noise, we would have found that minimizing  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_1$  was optimal.



$$Pr(\eta = z) \sim$$

Laplace noise has “heavier tails”, meaning that it results in more outliers.

**This is a completely different justification for  $\ell_1$  loss.**

# Bayesian regularization

We can add another layer of probabilistic modeling by also assuming  $\beta$  is random and comes from some distribution, which encodes our prior belief on what the parameters are.

Maximum a posteriori (MAP estimation):

$$\Pr(\beta \mid \mathbf{X}, \mathbf{y}) = \frac{\Pr(\mathbf{X}, \mathbf{y} \mid \beta) \Pr(\beta)}{\Pr(\mathbf{X}, \mathbf{y})}.$$

Assume values in  $\beta = [\beta_1, \dots, \beta_d]$  come from some distribution.

- **Common model:** Each  $\beta_i$  drawn from  $N(0, \gamma^2)$ , i.e. normally distributed, independent.
- Encodes a belief that we are unlikely to see models with very large coefficients.

**Goal:** choose  $\beta$  to maximize:

$$\Pr(\beta \mid \mathbf{X}, \mathbf{y}) = \frac{\Pr(\mathbf{X}, \mathbf{y} \mid \beta) \Pr(\beta)}{\Pr(\mathbf{X}, \mathbf{y})}.$$

- We can still ignore the “evidence” term  $\Pr(\mathbf{X}, \mathbf{y})$  since it is a constant that does not depend on  $\beta$ .
- $\Pr(\beta) = \Pr(\beta_1) \cdot \Pr(\beta_2) \cdot \dots \cdot \Pr(\beta_d)$
- $\Pr(\beta) \sim$

# Bayesian regularization

Easier to work with the **log likelihood**:

$$\begin{aligned} & \arg \max_{\beta} \Pr(\mathbf{X}, \mathbf{y} \mid \beta) \cdot \Pr(\beta) \\ &= \arg \max_{\beta} \prod_{i=1}^n e^{-(y_i - \langle \mathbf{x}_i, \beta \rangle)^2 / 2\sigma^2} \cdot \prod_{i=1}^n e^{-(\beta_i)^2 / 2\gamma^2} \\ &= \arg \max_{\beta} \sum_{i=1}^n -(y_i - \langle \mathbf{x}_i, \beta \rangle)^2 / 2\sigma^2 + \sum_{i=1}^d -(\beta_i)^2 / 2\gamma^2 \\ &= \arg \min_{\beta} \sum_{i=1}^n (y_i - \langle \mathbf{x}_i, \beta \rangle)^2 + \frac{\sigma^2}{\gamma^2} \sum_{i=1}^d (\beta_i)^2 / \sigma^2. \end{aligned}$$

Choose  $\beta$  to minimize  $\|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \frac{\sigma^2}{\gamma^2} \|\beta\|_2^2$ .

**Completely different justification for ridge regularization!**

**Test your intuition:** What modeling assumption justifies LASSO regularization:  $\min \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1$ ?

**Modeling language**



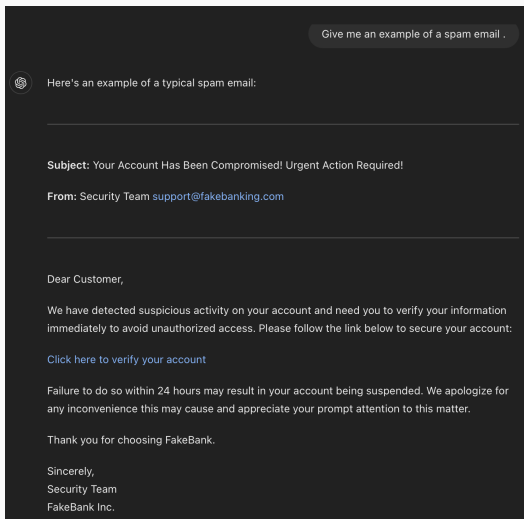


## Example

**Key idea behind generative ML:** Build a very good probabilistic model for your data. Use that model to generate realistic looking new data.

**Email example from our model:** *Keeps retaining in astro associated to no garden superconducting whistleblower on effusion eigenvalue jobs worker for car shortlist villa depictions fitness the easter veto devices expressed secondary user metal this administrative the do of to struct coffee online cde the open through requirement stamps you job g thus drop stations.*

How do we go from this to something more like what modern models can produce?



## How do we go from jumbled words to something more like what modern models can produce?

Dear Customer,

We have detected suspicious activity on your account and need you to verify your information immediately to avoid unauthorized access. Please follow the link below to secure your account:

[Click here to verify your account](#)

Failure to do so within 24 hours may result in your account being suspended. We apologize for any inconvenience this may cause and appreciate your prompt attention to this matter.

Thank you for choosing FakeBank.

Sincerely,  
Security Team  
FakeBank Inc.

---

### Warning signs of spam:

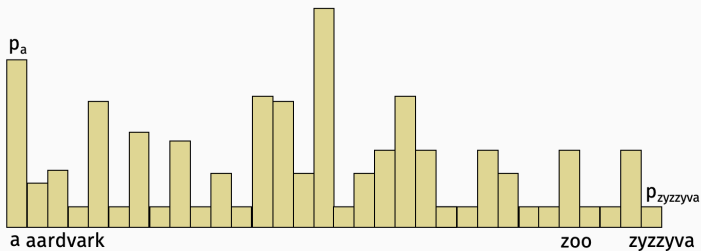
- **Urgency:** The message creates a sense of urgency ("urgent action required").
- **Unfamiliar sender:** The email address doesn't seem to match a legitimate institution.
- **Phishing link:** The link directs to a suspicious URL that doesn't belong to the claimed company.
- **Generic greeting:** It doesn't use your name, just "Dear Customer."
- **Threat:** It warns of a negative consequence if immediate action is not taken.

Always double-check such emails before clicking on any links or providing personal information!

# Autogressive models

**Main issue:** Our model lacks context!

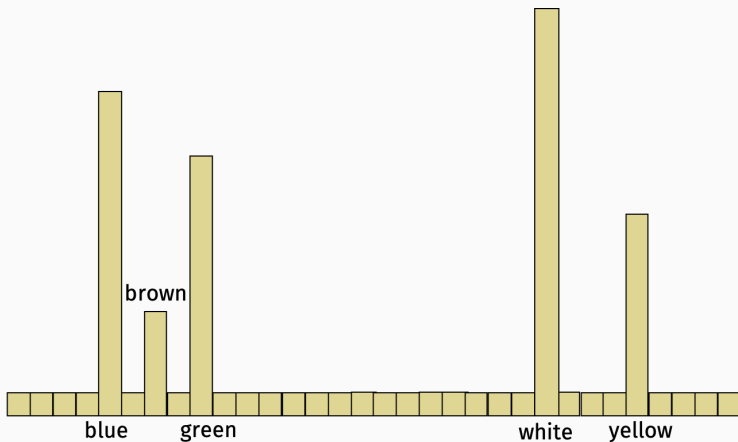
The color of the dress is \_\_\_\_\_.



# Autogressive models

**Main issue:** Our model lacks context!

The color of the dress is \_\_\_\_\_.



**Key idea:** Distribution that a word is chosen from should depend on previous words in the sentence/paragraph.

Consider generating a sentence with words  $x_1, x_2, \dots, x_n$ .

- Initialize the first word  $x_1$  of the sentence (e.g. at random).
- Choose  $x_2$  based on  $x_1$ .
- Choose  $x_3$  based on  $x_1, x_2, \dots$

Concretely, set  $x_i = w$  with probability:

$$P(x_i = w \mid x_{i-1}, x_{i-2}, \dots, x_1).$$

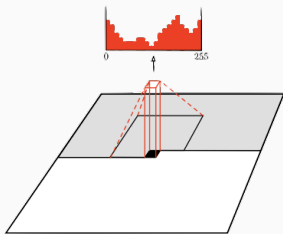
# Autoregressive models

Autoregressive model's generate text in order.

- How most humans write sentences, emails, short text.
- How the latest modern language models write text (e.g. the GPT family of models.)

**This is not the only approach to generative modeling, but it is one that works fairly well in practice, especially for text.**

Can also be used e.g. for images, but no longer state of the art.



## Limited lookback

**Key idea:** Distribution that a word is chosen from should depend on previous **k words** in the sentence/paragraph.  $k$  is a parameter that controls model complexity.

Consider generating a sentence with words  $x_1, x_2, \dots, x_n$ .

- Initialize the first  $k$  word  $x_1, \dots, x_k$  of the sentence (e.g. at random).
- Choose  $x_{k+1}$  based on  $x_1, \dots, x_k$ .
- Choose  $x_{k+2}$  based on  $x_2, \dots, x_{k+1}$ .
- Choose  $x_{k+3}$  based on  $x_3, \dots, x_{k+2}$ .
- ...

Set  $x_i = w$  with probability:

$$P(x_i = w \mid x_{i-1}, x_{i-2}, \dots, x_{i-k}).$$



## Limited lookback

Set  $x_i = w$  with probability:

$$P(x_i = w \mid x_{i-k}, \dots, x_{i-2}, x_{i-1}).$$

This probability can be tractably estimate from our data!

It is exactly the same as the probability of observing the  $k + 1$ -gram  $[x_{i-k}, \dots, x_{i-2}, x_{i-1}, w]$ .

### Training:

- For corpus of text, collect all  $k + 1$ -grams and record their frequency.

### Prediction:

- At step  $i$ , sample from the subset of  $k + 1$  grams starting with  $[x_{i-k}, \dots, x_{i-2}, x_{i-1}]$ , with probability proportional to their frequency.

## Example

The color of the dress is \_\_\_\_\_.

- Reasonable completions for  $k = 2$ :
  
- Reasonable completions for  $k = 5$ :

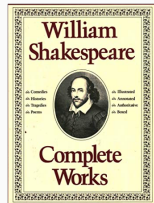
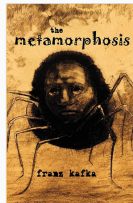
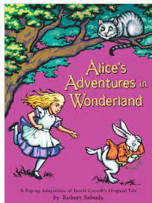
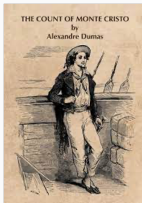
## Lab 3

Credit to Raphael Meyer who was a Ph.D. student at CSE Tandon, currently a postdoc at Caltech.



## Lab 3

- Train model on free books from Project Gutenberg.

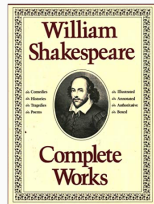
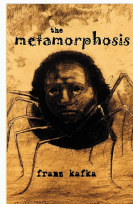
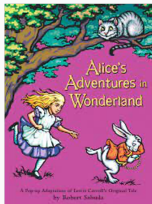
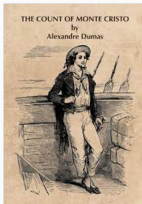


- Evaluate effect of changing  $k$ . Tradeoff between better performance and more “copying” from the course text.

*Virtue itself of vice must pardon beg,  
Yea, curb and woo for leave  
to do him good, She shall undo her credit with the judge, or own  
great place, Could fetch your brother from the angry law; do no  
stain to your own souls so blind That you will clear yourself from  
all suspense.*

## Lab 3

- Train model on free books from Project Gutenberg.



- Evaluate effect of changing  $k$ . Tradeoff between better performance and more “copying” from the source text.

*During this time, Madame Morrel had told her all,—‘Giovanni,’ said she, ‘you should have brought this child with you; we would have replaced the parents it has lost, have called it Benedetto, and then, in a loyal duel, and not in Arabia, and in France eternal friendships are as rare as the custom of doing when saying “Yes.” “Good; he accepts,” said Monte Cristo.*